

Machine Learning Solutions for Forward and Inverse Problems Based on Multiscale Modeling

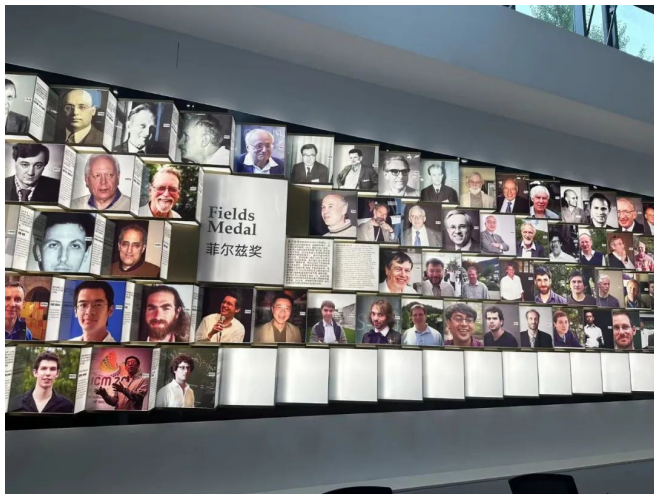
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Beijing Institute of Mathematical Sciences and Applications (BIMSA)



Research Direction

■ Scientific Machine Learning

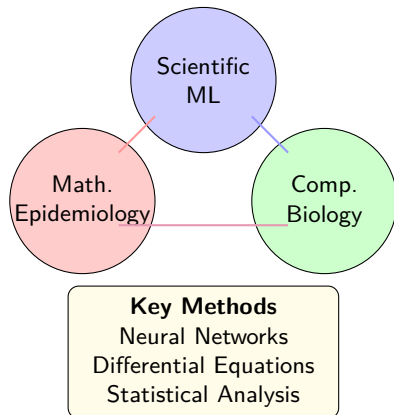
- Inverse problems
- PINN
- Maximum entropy Methods

■ Computational Biology

- Protein aggregation
- Complex biochemical reaction system

■ Mathematical Epidemiology

- COVID-19 modeling
- Control measure analysis



Selected Publications:

Entropy Structure Informed Learning for Solving Inverse Problems (2023)
When Machine Learning Meets Multiscale Modeling (2020)

Contents

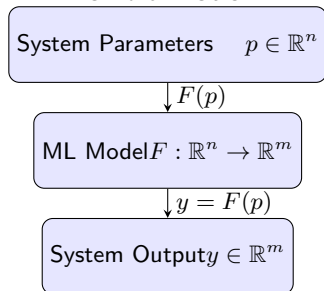
- 1 Revealing hidden dynamics from time-series data by ODENet
 - The structure of ODENet
 - Numerical experiments
 - Summary of Part One

- 2 Research on Multi-scale Modeling Based on Machine Learning
 - Model of cellular differentiation
 - Gene regulatory networks with negative feedback
 - Summary of Part Two

Revealing hidden dynamics from time-series data by ODENet

Forward Problem and Inverse Problem in Machine Learning

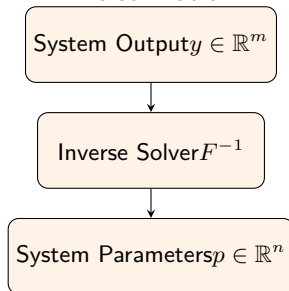
Forward Problem



Forward Problem: $y = F(p)$

- F : Forward operator
- $p \in \mathbb{R}^n$: Parameters
- $y \in \mathbb{R}^m$: Output
- Applications: Image generation; Physics simulation

Inverse Problem



Inverse Problem:

$$p = \arg \min_p \|F(p) - y\|^2 + \lambda R(p)$$

- Often ill-posed
- $R(p)$: Regularization
- λ : Reg. parameter
- Applications: System identification

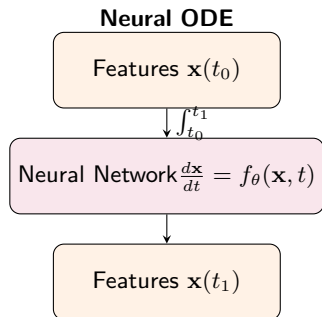
Overview of ML Methods

Category	Method	Key Features
Differential-based Methods	PINNs	▪ Physics-constrained loss ▪ Automatic differentiation ▪ Direct PDE solution
	SINDy	▪ Sparse regression ▪ Derivative discovery ▪ Interpretable models
	PDENet	▪ Differential operator learning ▪ Convolutional structure ▪ Finite difference inspired
Integral-based Methods	Neural ODEs	▪ Continuous-time models ▪ Memory efficiency ▪ Reversible architecture
	Hamiltonian Networks	▪ Energy conservation ▪ Symplectic structure
	OnsagerNet	▪ Thermodynamic consistency ▪ Variational principles ▪ Energy-based stability

Comparison of Approaches

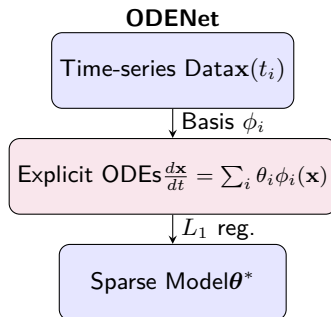
Aspect	Differential Methods	Integral Methods
Advantages	▪ Direct physics correspondence ▪ Simpler implementation ▪ Computational efficiency ▪ Local property handling	▪ Better numerical stability ▪ Noise robustness ▪ Conservation properties ▪ Global dynamics capture
Limitations	▪ Noise sensitivity ▪ Numerical instability risks ▪ Derivative quality dependent	▪ Higher computational cost ▪ Complex implementation ▪ Slower training process
Best Use Cases	▪ Clean data scenarios ▪ Fast computation needs ▪ Known physics equations ▪ Local dynamics focus	▪ Noisy data handling ▪ Long-term prediction ▪ Conservation law problems ▪ Global behavior analysis

From Neural ODE to ODENet



■ **Key Features:**

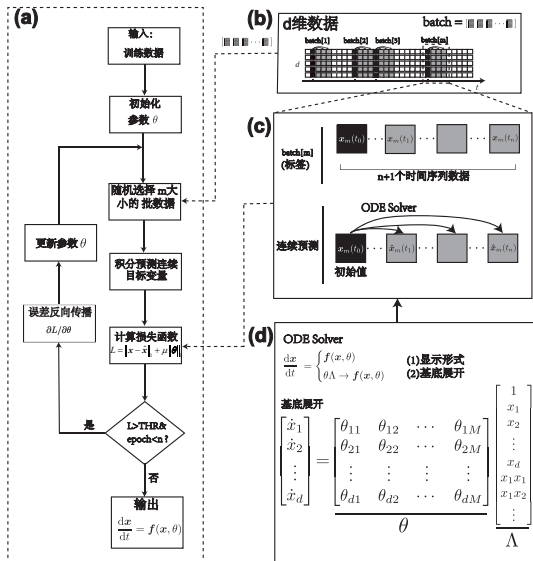
- Continuous-depth networks
- Implicit function learning
- Black-box dynamics
- Memory efficient



■ **Key Features:**

- Interpretable dynamics
- Explicit basis functions
- Sparse identification
- Noise robust

Consider a d -dimensional dynamical system $\dot{f}(x; \Theta) = \Theta \Phi(x)$

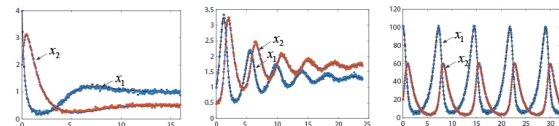


Lotka-Volterra(LV) dynamics

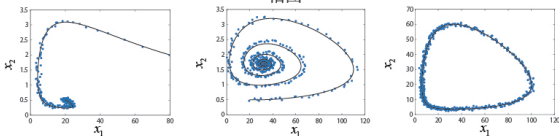
$$\begin{aligned}\frac{dx_1}{dt} &= C_{11}x_1 + C_{12}x_1x_2 + C_{13}x_1^2, \\ \frac{dx_2}{dt} &= C_{21}x_2 + C_{22}x_1x_2 + C_{23}x_2^2.\end{aligned}\tag{1}$$

- **Experimental data:** Different types of dynamic time trajectories when simulating equation (1) (with 1% or 10% white noise);
- **Basic settings:** Using complete polynomial Λ approximates $f(x)$.

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相图



LV Equation		Parameters					
		C_{11}	C_{12}	C_{13}	C_{21}	C_{22}	C_{23}
Overdamped Case 1%Noise	Model	1.5	-1	-1	-1	1	0
	ODENet	1.49	-9.87×10^{-1}	-9.94×10^{-1}	-1.00	9.95×10^{-1}	0
	SINDy#1($\Delta t = 0.01$)	—	—	—	—	—	—
	SINDy#2($\Delta t = 0.001$)	1.50	-1.00	-1.00	-1.00	1.00	0
Spiral Case 1%Noise	Model	2	-1.1	-0.1	-1	-0.1	0.9
	ODENet	1.99	-1.11	-9.91×10^{-2}	-9.87×10^{-1}	-1.06×10^{-1}	9.02×10^{-1}
	SINDy#1	—	—	—	—	—	—
	SINDy#2	2.18	-8.54×10^{-1}	-1.04×10^{-1}	-9.89×10^{-1}	-1.01×10^{-1}	8.65×10^{-1}
Limit cycle case 1%Noise	Model	1	-0.05	0	-1	0.03	0
	ODENet	9.97×10^{-1}	-4.98×10^{-2}	0	-1.00	2.99×10^{-2}	0
	SINDy#1	1.00	-4.98×10^{-2}	0	-9.98×10^{-1}	3.01×10^{-2}	0
	SINDy#2	1.01	-4.99×10^{-2}	0	-9.95×10^{-1}	3.00×10^{-2}	0
Limit cycle case 10%Noise	Model	1	-0.05	0	-1	0.03	0
	ODENet	9.69×10^{-1}	-4.89×10^{-2}	0	-9.82×10^{-1}	2.96×10^{-2}	0
	SINDy#1	9.83×10^{-1}	-5.04×10^{-2}	0	-1.00	3.02×10^{-2}	0
	SINDy#2	1.00	-4.99×10^{-2}	0	-9.97×10^{-1}	3.00×10^{-2}	0

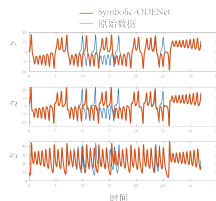
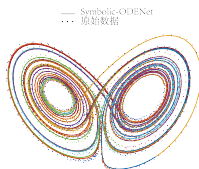
The impact of data sampling time step on results

Limit cycle with 1% noise		Parameters					
		C_{11}	C_{12}	C_{13}	C_{21}	C_{22}	C_{23}
Model		1	-0.05	0	-1	0.03	0
Methods	Δt						
ODENet	0.001	9.93×10^{-1}	-5.00×10^{-2}	0	-1.00	3.00×10^{-2}	0
SINDy	0.001	1.01	-4.99×10^{-2}	0	-9.95×10^{-1}	3.00×10^{-2}	0
ODENet	0.01	9.97×10^{-1}	-4.98×10^{-2}	0	-1.00	2.99×10^{-2}	0
SINDy	0.01	1.00	-4.98×10^{-2}	0	-9.98×10^{-1}	3.01×10^{-2}	0
ODENet	0.1	9.82×10^{-1}	-4.94×10^{-2}	0	-1.00	3.01×10^{-2}	0
SINDy	0.1	5.87×10^{-1}	-5.55×10^{-2}	0	0	1.83×10^{-1}	0
ODENet	0.5	9.90×10^{-1}	-4.99×10^{-2}	0	-1.02	3.08×10^{-2}	0
SINDy	0.5	—	—	—	—	—	—

- Compared to differential-based methods, integral-based methods are more adaptable to data sampled with large time steps.

Lorenz Equation in Chaotic Regime

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x), \\ \frac{dy}{dt} &= x(\rho - z) - y, \\ \frac{dz}{dt} &= xy - \beta z,\end{aligned}$$



Method	Parameters						
	C_{11}	C_{12}	C_{21}	C_{22}	C_{23}	C_{31}	C_{32}
Model	-10	10	28	-1	-1	-8/3	1
ODENet	-9.989	9.982	28.02	-1.008	-1.000	-2.667	1.000
SINDy	-9.899	9.976	26.72	-0.397	-0.975	-2.447	1.000

- Only 7 non-zero parameters shown (23 others are zero for ODENet)
- SINDy has an extra term -5.115 in third equation

Learn big noise in ODENet

- Noisy time series data:

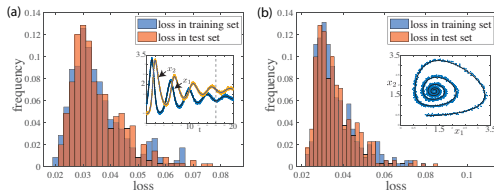
$$\mathbf{e}(t) = \epsilon \|\mathbf{y}\|_{\infty} \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(0, 1)$$

- ODENet governing equation:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}; \boldsymbol{\theta}), \quad \mathbf{x}(t) = \mathbf{y}(t) - \mathbf{e}(t)$$

- Loss function with noise parameters:

$$L = \|\hat{\mathbf{x}} + \hat{\mathbf{e}} - \mathbf{y}\|_1 + \mu \|\boldsymbol{\theta}\|_1$$



Summary of Part One

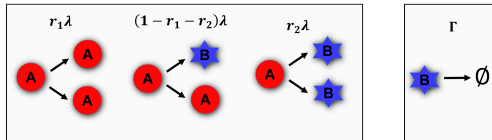
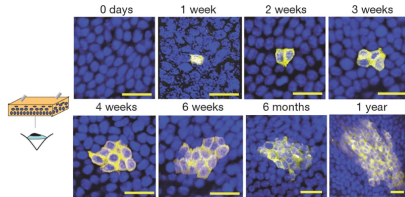
The work in Part One¹ can be summarized in the following four aspects:

1. Capable of handling non-uniform spacing and high signal-to-noise ratio data;
2. Adaptable to data with relatively large sampling time steps;
3. Explicitly derives interpretable and concise models with fewer parameters;
4. Highly flexible network structure that can readily incorporate new methods.

¹Pipi Hu et al. "Revealing hidden dynamics from time-series data by ODENet". In: *Journal of Computational Physics* 461 (2022), p. 111203.

Research on Multi-scale Modeling Based on Machine Learning

Model of cellular differentiation



Clayton et al., Nature, 2007, 446(7132): 185-189.

Master Equation for A/B-Type EPC Population Dynamics

Defining $P_{n_A, n_B}(t)$ as the probability that a labelled clone involves n_A A-type and n_B B-type EPCs at time t after induction, its timeevolution is governed by the Master equation:

$$\begin{aligned} \frac{dP_{n_A, n_B}}{dt} = & \lambda \left\{ r \left[(n_A - 1)P_{n_A-1, n_B} + (n_A + 1)P_{n_A+1, n_B-2} \right] + \right. \\ & \left. (1 - 2r)n_A P_{n_A, n_B-1} - n_A P_{n_A, n_B} \right\} + \\ & \Gamma \left[(n_B + 1)P_{n_A, n_B+1} - n_B P_{n_A, n_B} \right] \end{aligned} \quad (2)$$

subject to the initial condition: $P_{n_A, n_B}(0) = \rho \delta_{n_A, 1} \delta_{n_B, 0} + (1 - \rho) \delta_{n_A, 0} \delta_{n_B, 1}$

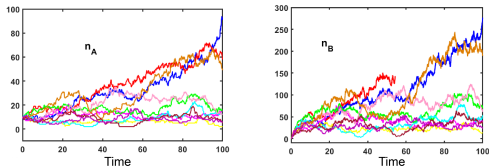


Figure: Gillespie algorithm (GA) generates stochastic orbital data.

Moment-closure equations in Kurtz's limit

From CMEs to MAEs

- Mean density of species:

$$c_i = \sum_n V^{-1} n_i p(n)$$

- Evolution equation:

$$\begin{aligned} \frac{d}{dt} \sum_n V^{-1} n_i p(n) &= \sum_{j=1}^M \sum_n V^{-1} n_i [p(n - u_j) \Phi_j(n - u_j) - p(n) \Phi_j(n)] \\ &= \sum_{j=1}^M u_{ij} \left[\sum_n V^{-1} p(n) \Phi_j(n) \right] \end{aligned}$$

Kurtz's Limit Process

- Take limit: $V \rightarrow +\infty$, $n \rightarrow +\infty$ while keeping $V^{-1}n$ finite
- Propensity scaling: $V^{-1}\Phi_j(n) = \phi_j(V^{-1}n) + O(V^{-1})$
- Results in Mass-Action Equations:

$$\frac{d}{dt} c_i(t) = \sum_{j=1}^M (\nu'_{ij} - \nu_{ij}) \phi_j(c)$$

Moment-closure equations in Kurtz's limit

Significance

- Reduces computational complexity
- Bridges mesoscopic and macroscopic scales
- Enables efficient modeling of large-scale chemical systems

*Rigorously justified by Kurtz's theorem (1970s)

First-order moment equation (mean equation)

- When there are many chemical reactions or reactants, computational simulation of the chemical master equation often encounters time-consuming problems; therefore, the **moment closure method** is used to simplify the chemical master equation;
- Define the first-order moment of $(n_A, n_B)^T$

$$\langle n_A \rangle = \sum_{n_A, n_B} n_A p(n_A, n_B), \quad \langle n_B \rangle = \sum_{n_A, n_B} n_B p(n_A, n_B). \quad (3)$$

- The time evolution equations for the ensemble average number of cells $\langle n_A \rangle$ of type A and the ensemble average number of cells $\langle n_B \rangle$ of type B

$$\begin{aligned} \frac{d \langle n_A \rangle}{dt} &= \lambda (r_1 - r_2) \langle n_A \rangle, \\ \frac{d \langle n_B \rangle}{dt} &= \lambda (1 - r_1 + r_2) \langle n_A \rangle - \Gamma \langle n_B \rangle. \end{aligned} \quad (4)$$

Determining Chemical Reaction Types

- Determine the types of chemical reactions between two reactants, the reaction orders, and reaction rate parameters.
- Here we consider chemical reactions up to second order, corresponding to combinations of chemical reactants $A, B, A + A, A + B, B + B$

$$\begin{cases} \frac{d\langle n_A \rangle}{dt} = \alpha_{11} \langle n_A \rangle + \alpha_{12} \langle n_B \rangle + \alpha_{13} \langle n_A \rangle^2 + \alpha_{14} \langle n_A \rangle \langle n_B \rangle + \alpha_{15} \langle n_B \rangle^2, \\ \frac{d\langle n_B \rangle}{dt} = \alpha_{21} \langle n_A \rangle + \alpha_{22} \langle n_B \rangle + \alpha_{23} \langle n_A \rangle^2 + \alpha_{24} \langle n_A \rangle \langle n_B \rangle + \alpha_{25} \langle n_B \rangle^2. \end{cases} \quad (5)$$

Parameters	α_{11}	α_{21}	α_{22}
True value	0.0079	1.0921	-0.3100
ODENet	0.0079	1.0903	-0.3094
Relative error	0.00%	0.16%	0.19%
SINDy	0.0082	1.1030	-0.3131
Relative error	3.80%	1.00%	1%

- Since all coefficients in front of the quadratic terms in equation (5) have been sparsified, we can conclude that only first-order reactions exist in the current system.
- Two first-order reactions $A \xrightarrow{k_1} \phi$ and $A \xrightarrow{k_2} B$ could also exist in principle, which means that the probability distributions of the numbers of cells A and B satisfy the following chemical master equation

$$\begin{aligned} \frac{d}{dt}p(n_A, n_B) = & k_1 [(n_A + 1)p(n_A + 1, n_B) - n_A p(n_A, n_B)] \\ & + k_2 [(n_A + 1)p(n_A + 1, n_B - 1) - n_A p(n_A, n_B)] \\ & + \lambda [r_1(n_A - 1)p(n_A - 1, n_B) + r_2(n_A + 1)p(n_A + 1, n_B - 2) \\ & + (1 - r_1 - r_2)n_A p(n_A, n_B - 1) - n_A p(n_A, n_B)] \\ & + \Gamma [(n_B + 1)p(n_A, n_B + 1) - n_B p(n_A, n_B)]. \end{aligned} \quad (6)$$

- Currently, we do not have exact knowledge of these six reaction rate constants k_1 , k_2 , r_1 , r_2 , λ and Γ .

Second-order moment equations (Variance equations)

$$\left\{ \begin{array}{l} \frac{d\langle n_A \rangle}{dt} = \alpha_{11} \langle n_A \rangle, \frac{d\langle n_B \rangle}{dt} = \alpha_{21} \langle n_A \rangle + \alpha_{22} \langle n_B \rangle, \\ \frac{dV_A}{dt} = \beta_{11} \langle n_A \rangle + \beta_{12} V_A, \\ \frac{dV_B}{dt} = \beta_{21} \langle n_A \rangle + \beta_{22} \langle n_B \rangle + \beta_{23} V_B + \beta_{24} Cov, \\ \frac{dCov}{dt} = \beta_{31} \langle n_A \rangle + \beta_{32} V_A + \beta_{33} Cov, \end{array} \right. \quad (7)$$

$V_A = \langle n_A^2 \rangle - \langle n_A \rangle^2$, $V_B = \langle n_B^2 \rangle - \langle n_B \rangle^2$ and $Cov = \langle n_A n_B \rangle - \langle n_A \rangle \langle n_B \rangle$
 Furthermore, we have the relationship between moment equation coefficients and chemical reaction rate parameters

$$\begin{aligned} \alpha_{11} &= (r_1 - r_2)\lambda - k_1 - k_2, \alpha_{21} = (1 - r_1 + r_2)\lambda + k_2, \alpha_{22} = -\Gamma, \\ \beta_{11} &= k_1 + k_2 + (r_1 + r_2)\lambda, \beta_{12} = -2(k_1 + k_2) + 2(r_1 - r_2)\lambda, \\ \beta_{21} &= k_2 + (1 - r_1 + 3r_2)\lambda, \beta_{22} = \Gamma, \beta_{23} = -2\Gamma, \beta_{24} = 2((1 - r_1 + r_2)\lambda + k_2), \\ \beta_{31} &= -(2r_2\lambda + k_2), \beta_{32} = (1 - r_1 + r_2)\lambda + k_2, \beta_{33} = (r_1 - r_2)\lambda - k_1 - k_2 - \Gamma. \end{aligned} \quad (8)$$

ODENet learning second-order moment equations

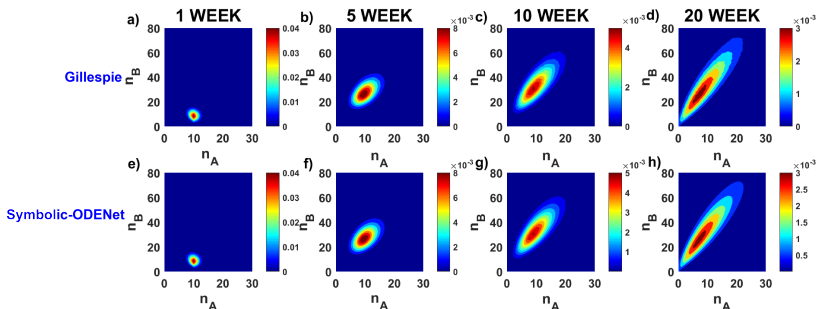
Parameters	β_{11}	β_{12}	β_{21}	β_{22}	β_{31}	β_{32}	β_{33}
	0.1760	0.0158	1.2602	0.3100	-0.1681	1.0921	-0.3021
	0.1756	0.0159	1.2678	0.3063	-0.1626	1.0807	-0.2984
	0.23%	0.63%	0.60%	1.19%	3.27%	1.04%	1.22%

$k_1, k_2, r_1, r_2, \lambda, \Gamma, \alpha, \beta$

$$\underbrace{\begin{bmatrix} 1 & -1 & -1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 2 & -2 & -2 & -2 & 0 & 0 \\ -1 & 3 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 1 & 0 \\ 1 & -1 & -1 & -1 & 0 & -1 \end{bmatrix}}_V \underbrace{\begin{bmatrix} r_1 \lambda \\ r_2 \lambda \\ k_1 \\ k_2 \\ \lambda \\ \Gamma \end{bmatrix}}_{\hat{u}} = \underbrace{\begin{bmatrix} \alpha_{11} \\ \alpha_{21} \\ -\alpha_{22} \\ \beta_{11} \\ \beta_{12} \\ \beta_{21} \\ \beta_{22} \\ -\beta_{31} \\ \beta_{32} \\ \beta_{33} \end{bmatrix}}_{\hat{b}}. \quad (9)$$

Reconstruction of Chemical Master Equation (CME)

Chemical reaction rate parameters	k_1	k_2	r_1	r_2	λ	Γ
True	0	0	0.0836	0.0764	1.1000	0.3100
Predicted	0.0123	-0.0197	0.0831	0.0824	1.1059	0.3074
Relative error	—	—	0.60%	7.85%	0.54%	0.84%



Basic Architecture of MEP-Net

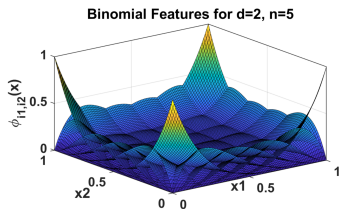
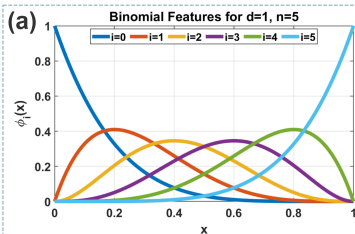
Loss Function Design:

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{constraint}}(\theta) + \lambda \mathcal{L}_{\text{entropy}}(\theta)$$

where

$$\mathcal{L}_{\text{constraint}}(\theta) = \frac{1}{T} \sum_{k=1}^T \sum_{i=1}^M [\mathbb{E}_{\hat{p}_{\theta}}[f_i] - \tilde{f}_i]^2$$

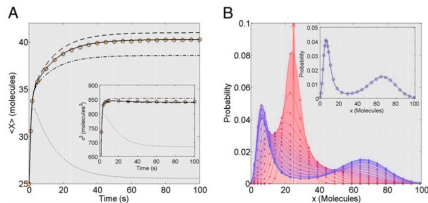
$$\mathcal{L}_{\text{entropy}}(\theta) = \frac{1}{T} \sum_{k=1}^T \mathcal{S}[\hat{p}_{\theta}]$$



Chemical Master Equation - Schlögl Model

The molecular number of X can exhibit bi-stability, characterized by two distinct stable states². **Model Evolution:**

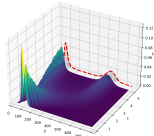
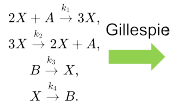
$$\begin{aligned} \frac{dp(n, t)}{dt} = & [k'_1(n-1)(n-2) + k'_3]p(n-1, t) \\ & - [k'_1n(n-1) + k'_3]p(n, t) \\ & + [k_2(n+1)n(n-1) + k_4(n+1)]p(n+1, t) \\ & - [k_2n(n-1)(n-2) + k_4n]p(n, t) \end{aligned}$$



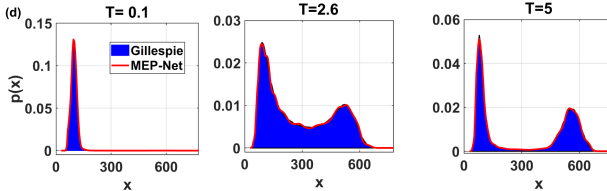
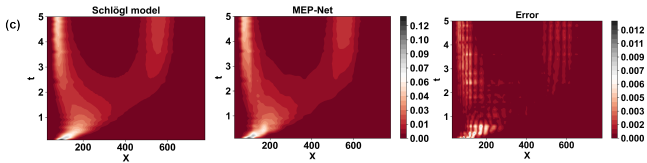
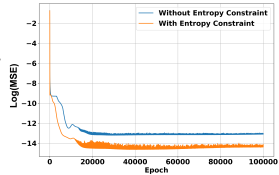
²Patrick Smadbeck and Yiannis N Kaznessis. "A closure scheme for chemical master equations". In: *Proceedings of the National Academy of Sciences* 110.35 (2013), pp. 14261–14265.

Results of Schlögl Model

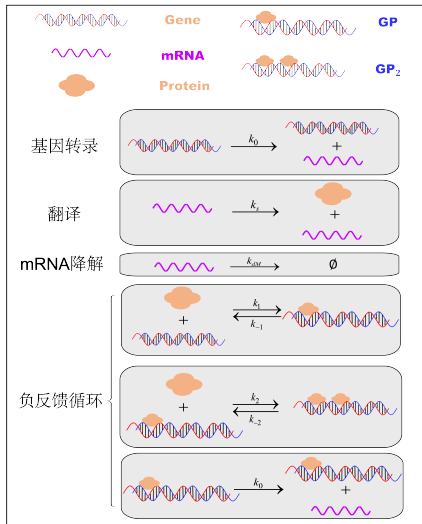
(a)



(b)



Gene regulatory networks with negative feedback



Basic model

- Chemical mass action equations describe gene networks,

$$\begin{cases} \frac{d}{dt}c_P = k_s c_M - k_1 c_G c_P + k_{-1} c_{GP} - k_2 c_P c_{GP} + k_{-2} c_{GP_2}, \\ \frac{d}{dt}c_M = k_0 c_G + k_0 c_{GP} - k_{dM} c_M, \\ \frac{d}{dt}c_G = -k_1 c_G c_P + k_{-1} c_{GP}, \\ \frac{d}{dt}c_{GP} = k_1 c_G c_P - k_{-1} c_{GP} - k_2 c_P c_{GP} + k_{-2} c_{GP_2}, \\ \frac{d}{dt}c_{GP_2} = k_2 c_P c_{GP} - k_{-2} c_{GP_2}. \end{cases} \quad (10)$$

- $c_G + c_{GP} + c_{GP_2} = c_{total}$ satisfies the law of mass conservation. The initial conditions are set as $c_G(0) = c_{GP}(0) = c_{GP_2}(0) = 0.01$, $c_P(0) = 0$, $c_M(0) = 5$.

Chemical reaction systems with time scale separation

- $k_1 = 3, k_{-1} = 2.4, k_2 = 9, k_{-2} = 6, k_0 = 0.05, k_s = 0.01, k_{dm} = 0.01$
- Let me create a chemical reaction system with **time scale separation**:

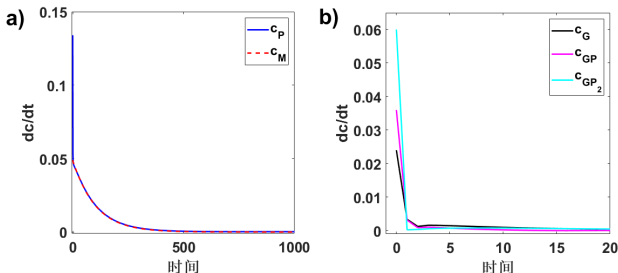


Figure: The derivative of a) slow variables and b) fast variables with respect to time.

ODENet for model reduction

- **Learning objective:** Automatic model order reduction for equation (10).
- Inspired by classical Michaelis-Menten kinetics results, we assume that the three fast variables c_G, c_{GP}, c_{GP_2} are expressed as rational functions, where both numerator and denominator are polynomials in c_P ,

$$H(c_G) = \frac{\Omega_1}{\Omega}, H(c_{GP}) = \frac{\Omega_2}{\Omega}, H(c_{GP_2}) = \frac{\Omega_3}{\Omega}. \quad (11)$$

- where $\Omega = \beta_1 + \beta_2 c_P + \beta_3 c_P^2$, $\Omega_1 = \alpha_{11} + \alpha_{21} c_P + \alpha_{31} c_P^2$, $\Omega_2 = \alpha_{12} + \alpha_{22} c_P + \alpha_{32} c_P^2$, $\Omega_3 = \alpha_{13} + \alpha_{23} c_P + \alpha_{33} c_P^2$.
- Since the last three equations in (10) do not contain terms with c_M , c_M does not appear in the rational functions.

$$\begin{cases} \frac{d}{dt} c_P = k_s c_M - \frac{k_1 c_P H(c_G) + k_{-1} H(c_{GP}) - k_2 c_P H(c_{GP}) + k_{-2} H(c_{GP_2})}{\Omega}, \\ \frac{d}{dt} c_M = -k_{dM} c_M + k_0 H(c_G) + k_0 H(c_{GP}). \end{cases} \quad (12)$$

Model parameters prediction

Parameters	QSSA	ODENet	Relative error	SINDy	Relative error
β_1	0.53	0.54	1.89%	0.44	16.98%
β_2	0.67	0.66	1.49%	0.72	7.46%
β_3	1.00	1.00	—	1.00	—
α_{11}	0.0159	0.0163	2.52%	0.0160	0.63%
α_{21}	0	0	—	0	—
α_{31}	0	0	—	0	—
α_{12}	0	0	—	0	—
α_{22}	0.0201	0.0200	0.50%	0.0201	0%
α_{32}	0	0	—	0	—
α_{13}	0	0	—	0	—
α_{23}	0	0	—	0	—
α_{33}	0.0300	0.0300	0%	0.0302	0.67%

Summary of Part Two

The work in Part Two³ can be summarized in three aspects:

- Inference of chemical reaction networks;
- Multi-scale modeling assisted machine learning;
- Automatic model reduction of systems with time scale separation using machine learning.

³Wuyue Yang et al. "When machine learning meets multiscale modeling in chemical reactions". In: *The Journal of Chemical Physics* 153.9 (2020), p. 094117.

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Welcome excellent students to join our team!

Thank you!